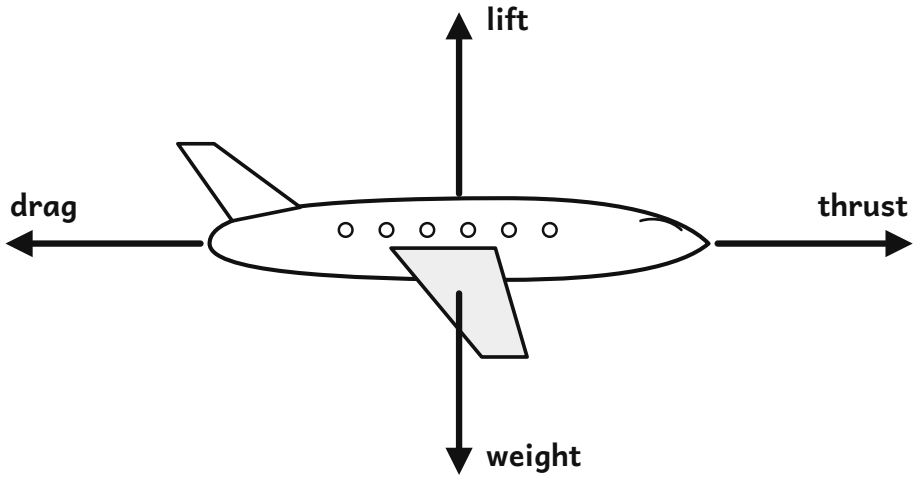
PHYSICAL AND CHEMICAL SCIENCE

Floating and Flying

Nature's floaters and flyers, the four forces of flight, upthrust, density and the properties of air



In this pack

Learn it: Floating and flying ★☆☆
Fact check: Forces for floating and flying ★☆☆
Think it through: Why do things float and fly? ★★☆
Investigate: Foil boats: which shape holds most? ★★☆
Real-world case study: Safe on the water: boats and launches ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Floating and flying

★☆☆ Grade 4

Name: Class: Date:

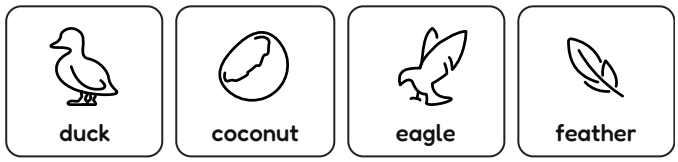
Learning goals

- Describe features of plants and animals that help them float or fly, and link them to human-made floaters and flyers.
- Label the four forces of flight and explain how they work together.
- Explain floating using weight, upthrust, displaced water and density, and describe the properties of air.

In a country of rivers, boats are everywhere: wooden *nouka*, river launches, fishing trawlers and huge cargo ships. Above them, kites, birds and aeroplanes fill the sky. How does a heavy steel ship float? How does a bird stay up? The answers are all about **forces**.

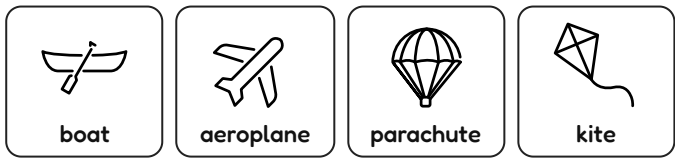
Nature's floaters and flyers

Living thing	Feature	Helps it to...
duck	oily, waterproof feathers trap air; webbed feet	float and paddle
fish	an air bag inside (swim bladder)	float at the right depth
water hyacinth	stalks full of air	float on rivers
coconut	air-filled husk	float across the sea to new islands
bird	light hollow bones, feathers, curved wings	fly
silk-cotton (*shimul*) seed	fluffy fibres	drift on the wind

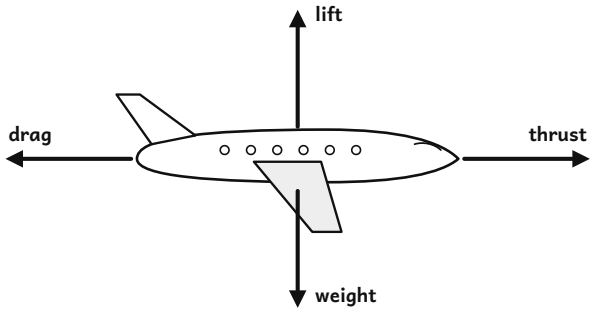


Copying nature

People have copied these features. **Life jackets** trap air like a duck's feathers. **Boats** have wide hollow hulls. **Aeroplane wings** are curved like a bird's. A **parachute** catches the air like a fluffy seed. **Kites** and gliders use the wind like soaring eagles.



The four forces of flight

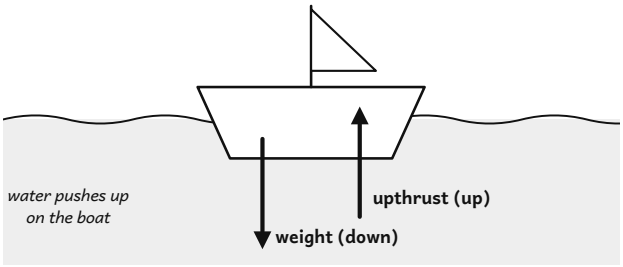


The four forces on a plane.

- **Lift** pushes the plane **up**. It is made by air flowing over and under the curved wings.
- **Weight** (gravity) pulls the plane **down**.
- **Thrust** pushes the plane **forward**. It comes from the engines (or a bird flapping its wings).
- **Drag** (air resistance) pulls the plane **back**.

To take off, **lift must be bigger than weight** and **thrust bigger than drag**. When the forces are balanced, the plane flies level at a steady speed.

How things float



Forces on a floating boat.

When you put an object in water, it **pushes some water out of the way** (it **displaces** water). The water pushes back up on the object. This upward push is called **upthrust** (or buoyancy). Gravity pulls the object down (its **weight**).

- If the upthrust can become **as big as the weight**, the object **floats**.
- If the weight is **bigger** than the biggest upthrust, the object **sinks**.

Density

Density tells us how heavy something is for its size. A stone is heavy for its size (dense), so it sinks. Wood is light for its size, so it floats. A steel ship floats because it is **hollow**: its shape is full of air, so the ship as a whole is light for its size and it displaces a lot of water.

The properties of air

- Air **takes up space** and **has mass**.
- Air can be **squashed** (compressed): we pump it into tyres and balls.
- Air **pushes** on things (air pressure) and **moves** (wind).

We use these properties in tyres, balloons, life jackets, kites, sails, windmills and parachutes.

Bangladesh connection

In winter, Old Dhaka celebrates **Shakrain** with hundreds of colourful kites flying from the rooftops. Kite fliers use the wind to give their kites lift, and the string's tension to steer them.

Key words

float	stay on the surface of a liquid
upthrust (buoyancy)	the upward push of water or air
displace	push out of the way
density	how heavy something is for its size
lift	the upward force on a wing
thrust	the forward force from engines or flapping
drag	air resistance that pulls a flyer back
weight	the pull of gravity

FACT CHECK · C5 · FLOATING AND FLYING

Forces for floating and flying

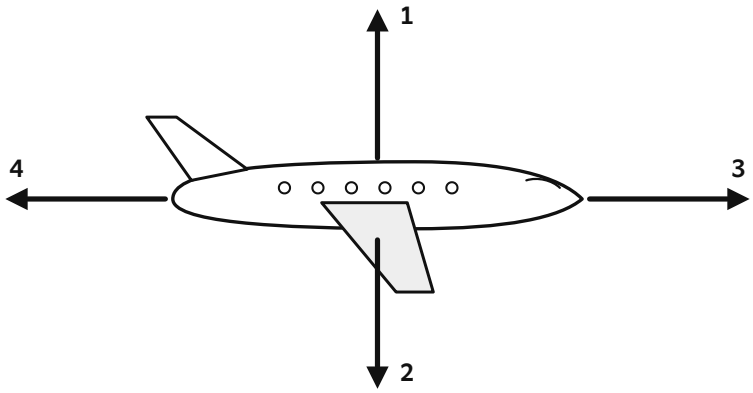
★☆☆
Grade 4

Name: Class: Date:

Learning goals

- Label the forces of flight.
- Sort floaters and sinkers.

1 Label the four forces of flight. [4]



1	2	3	4

2 Predict: float or sink? [8]

SORT THESE

a glass marble **a wooden stick** **a plastic duck** **a cork** **a coin** **a dry leaf** **a metal key** **a stone**

Floats	Sinks

3 **Match the human invention to the living thing it copies.** [4]

Draw a line to match each pair, or write the letter.

life jacket ●	● **A** duck feathers trap air
parachute ●	● **B** curved bird wing
aeroplane wing ●	● **C** fluffy seeds drifting on the wind
boat hull ●	● **D** floating coconut shell

4 **Fill in the gaps.** [3]

WORD BANK

smaller **gravity** **bigger** **upthrust** **squashed**

The upward push of water is called _____. A stone sinks because its weight is _____ than the upthrust. Air can be _____ into a tyre.

Why do things float and fly?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Explain floating with forces and density.
- Explain flight forces.

1 **A steel nail sinks, but a steel ship floats. Explain why.** [3]

2 **Why do life jackets help you float?** [2]

3 **A plane is taking off. Which forces must be bigger? Tick two.** [2]

	True?
lift is bigger than weight	
weight is bigger than lift	
thrust is bigger than drag	
drag is bigger than thrust	

4 **Why does a parachute have such a big canopy?** [2]

5 **Draw a boat with arrows for weight and upthrust. Show what happens when the boat is overloaded.** [2]

Boat forces

INVESTIGATE · C5 · FLOATING AND FLYING

Foil boats: which shape holds most?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Design and test model floaters.
- Record results.
- Improve a design.

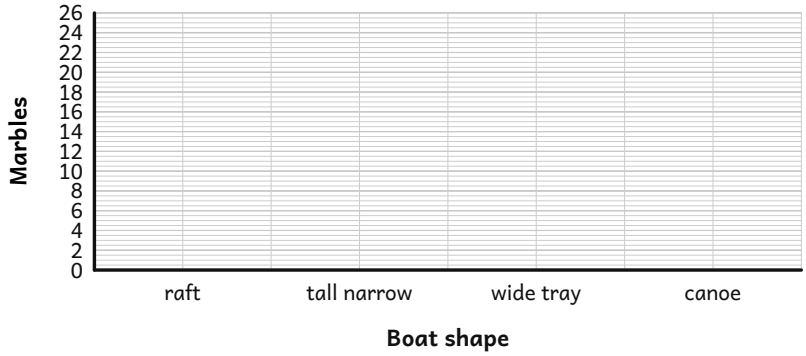
Class 4A made boats from the same size of aluminium foil (20 cm × 20 cm) in four shapes. They added marbles one at a time until each boat sank.

1 Complete the plan. [3]

Change	
Measure	
Keep the same	

2 Draw a bar chart. [3]

Boat shape	flat raft	tall narrow	wide low tray	canoe
Marbles before sinking	6	9	24	15



3 Which shape held the most? Explain using the word displace. [2]

.....

.....

4 How could you improve the best boat? [2]

.....

.....

InquiryMind
LEARN · PLAY · WONDER

Safe on the water: boats and launches

★★★
Grade 4

Name: Class: Date:

Learning goals

- Apply floating science to safety.
- Use data about loads.
- Suggest rules to save lives.

Science · Citizenship · Maths · Geography

Millions of people in Bangladesh travel by water every day: small *nouka*, ferries and big **launches** that carry hundreds of passengers between Dhaka and the south. When a boat is loaded, it sinks lower in the water. Every launch is allowed to carry only a certain number of people.

Sometimes, especially before holidays like Eid, launches are **overloaded**. The boat sits too low, waves can wash in, and it becomes unstable. In storms, overloaded boats can capsize. Safety rules say launches must not leave in bad weather, must carry life jackets and life buoys, and must not carry more people than allowed.

Based on river transport safety rules in Bangladesh. Data below are example values.

1 **A launch is allowed to carry 300 passengers (example data).** [3]

Trip	A	B	C	D
Passengers	250	300	420	280

a) Which trip was overloaded? By how many people?

b) Why is an overloaded boat more likely to sink?

2 **Many boats have a painted line on the side called a load line. What do you think it is for?** [2]

3 **What might each person say about overloading?** [3]

Person	View
a passenger going home for Eid	
the launch owner	
a river police officer	

4 **Design a safety poster for a launch terminal. Include two rules and a picture.** [2]

Safety poster

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

S	T	H	R	U	S	T	A	I	R
B	L	W	L	F	L	O	A	T	D
Y	U	E	I	D	B	S	W	O	R
O	A	I	F	V	V	I	U	W	A
J	N	G	T	C	F	N	L	S	G
O	P	H	C	O	K	K	L	Y	B
S	L	T	K	B	D	Y	I	J	S
B	O	A	T	F	K	G	P	L	E
U	U	P	T	H	R	U	S	T	V
O	P	B	T	L	H	B	V	L	D

air boat drag float lift sink thrust upthrust weight

2 Which force pulls an aeroplane backwards? [1]

- ◯ lift
- ◯ thrust
- ◯ drag
- ◯ weight

3 Why does a block of wood float? [1]

- ◯ it is heavy for its size
- ◯ it is light for its size
- ◯ it has no weight
- ◯ water pulls it up

4 Design a paper glider. Draw it and explain one change you could test to make it fly further. [3]

My glider

5 **Challenge. A ball of clay sinks, but the same clay shaped into a bowl floats. Explain.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
link animal features to human floaters and flyers	○	○	○
label the four forces of flight	○	○	○
explain upthrust and weight	○	○	○
explain floating using density and displacement	○	○	○
test and improve a model boat	○	○	○
explain why overloading boats is dangerous	○	○	○